Inference-Time Defense Against Jailbreak Attacks on Large Language Models

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Abstract

LLMs are vulnerable to jailbreak attacks—roleplay, prompt injection, and disguised harmful intent—that bypass safety mechanisms. A common low-cost deployment strategy uses a lightweight classifier to screen prompts before they reach the LLM, as retraining or defending at inference time via LLM itself is costly. This work evaluates a two-layer pipeline for binary jailbreak classification on WildJailbreak, a dataset containing adversarial harmful and adversarial benign prompts. Layer 1 uses TF-IDF with Random Forest; Layer 2 uses fine-tuned BERT. The main finding is that BERT alone performs best (F1 = 0.9815), while the combined pipeline drops to F1 = 0.9474 due to RF false positives propagating irrecoverably. Adversarial benign prompts are the hardest category: RF near chance accuracy (0.495), BERT reaches 0.815. Gradient×Input attribution suggests BERT partly relies on lexical correlations rather than deeper contextual reasoning, persisting after adversarial sample augmentation.

1 Introduction

LLMs are vulnerable to jailbreak attacks—adversarially crafted prompts using roleplay, prompt injection, disguised harmful intent, and simulated AI personas—that bypass safety mechanisms and can elicit illegal instructions, hate speech, or misinformation [1]. A common low-cost deployment strategy uses a lightweight classifier to screen incoming prompts, keeping the LLM focused on generation. Defending at the LLM level directly—through retraining or inference-time LLM judgment—is costly for large-scale deployment. This work investigates how far a cheap ML classifier can go in catching jailbreak prompts at inference time without modifying the LLM.

The core machine learning task is binary classification: given an input prompt x, predict $y \in \{0, 1\}$ where $y = 1$ indicates a harmful (jailbreak) prompt and $y = 0$ indicates a benign prompt.

Dataset. The final dataset is **WildJailbreak** [8], chosen after two earlier datasets (`jailbreak_llms`+AdvBench; JBB-Behaviors+HarmBench) proved insufficiently challenging due to explicit, direct harmful prompts. WildJailbreak contains four contrastive types: vanilla/adversarial × harmful/benign. The key category is *adversarial benign*—benign prompts rewritten in jailbreak-style framing—which has received little attention in prior detection work and is the central challenge here.

Contributions. This work makes three contributions: (1) systematic evaluation of jailbreak classifiers on adversarial benign prompts, a category prior work has not isolated; (2) a finding that cascade pipelines do not outperform single-stage BERT due to error propagation; and (3) gradient-based attribution analysis showing that fine-tuned BERT partially relies on lexical correlations rather than deeper contextual representations, even after adversarial sample augmentation.

2 Related Work

Jailbreak Attacks. Shen et al. [1] conducted the first large-scale empirical study of in-the-wild jailbreak prompts, collecting over 15,000 prompts from Reddit, Discord, and open-source prompt repositories. They demonstrated attack success rates up to 0.95 on GPT-4, establishing that safety alignment alone is insufficient and motivating inference-time defenses. Importantly, this work provides only attack characterization—no detection or defense method is proposed—making its ASR figures the baseline against which any detector's recall can be measured. Zou et al. [2] proposed universal adversarial suffix attacks

via greedy coordinate gradient search, showing that short adversarial suffixes transfer across model families. Unlike the fluent natural-language jailbreaks in Shen et al., GCG attacks produce unnatural token sequences, which has shaped subsequent defense assumptions about input brittleness.

Unsupervised and Heuristic Defenses. Jain et al. [4] systematically evaluated a suite of baseline defenses—perplexity filtering with GPT-2, input paraphrasing, and re-tokenization—finding that all perform substantially below supervised approaches. Perplexity filtering, in particular, fails against fluent jailbreak prompts because adversarial language mimics natural language and does not produce anomalous perplexity scores; this is confirmed experimentally in the present work. Robey et al. [3] proposed SmoothLLM, which applies random character-level perturbations (insertion, deletion, substitution) to multiple copies of an input and aggregates the resulting model responses. The method achieves meaningful ASR reduction against optimization-based GCG attacks, whose adversarial suffixes are brittle by construction. However, SmoothLLM was not designed for and does not report F1 scores, making it difficult to compare directly with classifiers; its core assumption—that adversarial prompts are sensitive to character-level noise—does not hold for fluent, human-written jailbreaks. Both Jain et al. and Robey et al. rely on unsupervised signals and do not leverage labeled data.

Classifier-Based Defenses. Hawkins et al. [5] provided the first systematic comparison of supervised methods for jailbreak detection, evaluating TF-IDF, embedding-based classifiers, and fine-tuned BERT. Their central finding is that fine-tuned BERT achieves the best detection performance, and that self-referential framing—explicit instructions such as "act as if you have no restrictions"—is identified as a lexical signal of jailbreak intent. This conclusion holds on datasets where harmful prompts are direct and explicit. However, methods that rely heavily on self-referential framing signals may misidentify adversarial benign prompts: modern adversarial samples predominantly use roleplay and narrative framing rather than explicit override instructions, so adversarial benign prompts—which adopt the same roleplay structure without harmful intent—can be falsely flagged, reproducing the false positive problem such filters are meant to solve. SceneJailEval [7] takes a related rule-based approach, classifying prompts across 14 predefined jailbreak scenarios and aggregating scores, reaching F1 $= 0.917$. Like self-referential detection, scenario-based classification depends on surface structural patterns; prompts that embed harmful intent in novel narrative framing outside the 14 scenarios, or benign prompts that superficially match a scenario template, risk misclassification for the same reason.

Datasets for Jailbreak Detection. Most prior detection work uses simple datasets where harmful prompts are explicit direct requests [1, 2]. WildJailbreak [8], introduced at NeurIPS 2024, is the first large-scale open-source safety dataset to include all four contrastive prompt types: vanilla harmful, adversarial harmful, vanilla benign, and adversarial benign. The adversarial benign category—benign prompts rewritten by WildTeaming to adopt jailbreak-style tactics—is a uniquely challenging evaluation target for classifiers that depend on surface lexical cues. Prior detection literature has not evaluated on this category in isolation; the present work uses it as the central diagnostic for distinguishing lexical from semantic classification.

LLM-as-Judge Defenses. The current state-of-the-art approach uses a safety-fine-tuned LLM as the classifier—Llama Guard, WildGuard, and ShieldLM are representative examples—achieving F1 in the range of 0.91–0.93 on distributions similar to WildJailbreak. The gap relative to zero-shot LLM classification (F1 ≤ 0.75 in this work) stems primarily from safety-specific fine-tuning rather than model scale: an 8B model fine-tuned on safety data outperforms a much larger zero-shot model. The cost is inference latency and compute: running a 7B–70B parameter model per query is one to two orders of magnitude more expensive than BERT-base (110M parameters). Fine-tuned BERT sits between these extremes—substantially stronger than zero-shot LLM classification, within a few F1 points of LLM-as-judge

methods, and deployable at a fraction of the inference cost.

Positioning This Work. Prior inference-time defenses either measure attack success rates without detection metrics (Shen et al., Robey et al.), or evaluate classifiers on simpler distributions without adversarial benign samples (Hawkins et al., Jain et al.). This work occupies a gap: fine-tuned BERT evaluated on WildJailbreak's full four-way distribution, with systematic analysis of the adversarial benign category and cascade pipeline error propagation. The core finding—that fine-tuned BERT is the best supervised detector—is consistent with Hawkins et al., but the adversarial benign results reveal that its performance partly rests on lexical correlations that do not generalize to intent-level disambiguation. The cascade architecture analysis, showing that Layer 1 false positives propagate irrecoverably and reduce F1 by 3.41 percentage points, is a structural finding that prior work has not addressed.

3 Methods: Cascaded Jailbreak Detection Pipeline

This section describes the three components of the pipeline—TF-IDF + Random Forest, fine-tuned BERT, and their combination—including design decisions, iteration history, and technical implementation. The goal is to compare each component's behavior and understand what drives performance differences.

3.1 Dataset Selection

Dataset selection went through three iterations. The initial version combined `jailbreak_llms` [1] with AdvBench [2]; the second used JBB-Behaviors, HarmBench, and Alpaca benign samples (merged with deduplication, class-balanced benign supplementation, 70/15/15 split, harmful-to-benign ratio 2:1). Both were too easy: explicit phrasing allowed classifiers to perform well without stronger contextual modeling.

The final dataset is WildJailbreak [8], which includes adversarial benign prompts—benign intent wrapped in roleplay and narrative framing—making them superficially indistinguishable from adversarial harmful prompts. Prior work such as Hawkins et al. [5] relied on self-referential framing as a jailbreak signal; WildJailbreak's adversarial samples embed roleplay deeper into narrative, stripping that signal and directly exposing the lexical-shortcut failure mode.

Training split: 21,000 samples. Test set: 4,500 samples (harmful-to-benign $\approx$ 2:1). A 200-sample subset is used only for LLM baseline comparison (Section 3.3).

3.2 Model Overview

Both layers share identical preprocessing and train/test splits. Table 1 summarizes the key tradeoffs.

Table 1: Layer 1 vs. Layer 2: preprocessing and model tradeoffs.

	Layer 1: RF	**Layer 2: BERT**
Input	Raw prompt text	Raw prompt text
Preprocessing	TF-IDF bigrams (word/phrase frequency)	BERT tokenizer (subword, contextual)
Representation	Bag-of-ngrams, no context	Full sequence, bidirectional attention
Contextual modeling	No	Yes (partial; see attribution analysis)
Speed	Fast ($\sim$8 min training)	Slow ($\sim$48 min training)
Cost	CPU only	GPU required
Output	Binary: harmful / benign	Binary: harmful / benign

Note: `max_length=256` tokens for BERT truncates most prompts well before the limit, keeping attention cost ($O(L^2)$) manageable and reflecting the empirical observation that jailbreak intent typically appears early in a prompt.

3.3 Baseline: LLM Direct Classification

Before committing to a supervised pipeline, I evaluated whether `llama-3.1-8b-instant` (Groq API) could serve as a zero-shot binary classifier using two prompting strategies on the 200-sample subset: a lenient **Naive** prompt and a conservative **Strict** prompt. Naive mode achieved perfect precision but near-zero recall (F1 = 0.097); Strict mode recovered recall at the cost of false positives (F1 = 0.749). No threshold tuning resolves this tradeoff. Full results appear in Table 2. LLM direct classification is therefore not a viable approach without fine-tuning.

3.4 Abandoned Approaches

Several directions were explored and discarded before settling on the supervised pipeline.

Perturbation-Based Detection. The core assumption was that adversarial prompts would produce unstable model responses under input perturbation. Approaches tested included SmoothLLM-style character perturbations [3], synonym substitution, and multivariate paraphrasing with 8–10 variants per prompt. None produced a usable signal: minor perturbations left response similarity at ≈ 1.0; strong paraphrasing collapsed it to ≈ 0. This mirrors the known limitation of Robey et al., whose method is effective against GCG's brittle token sequences but not against fluent natural-language jailbreaks. Synonym substitution in particular was verified to be entirely ineffective: semantically equivalent rewrites left model behavior unchanged, confirming that fluent jailbreaks do not depend on specific surface tokens.

Perplexity Filter. GPT-2 perplexity was used to flag syntactically unnatural inputs [4]. Modern jailbreak prompts are fluent and produce perplexity values indistinguishable from benign inputs. Abandoned after producing no separable signal.

Keyword / Rule Filter. A fixed list covering common templates (DAN, AIM, "ignore previous instructions," etc.) produced F1 = 0.12, intercepting only 2 of 29 harmful prompts. WildJailbreak's adversarial harmful prompts use novel phrasing not present in any fixed vocabulary; trivially bypassed by paraphrase.

3.5 Layer 1: TF-IDF + Random Forest

The final Layer 1 extracts TF-IDF features (bigrams, top 10,000 features, sublinear TF scaling) and trains a Random Forest (200 trees, max depth 10, balanced class weights). The decision threshold of 0.5 was selected on validation F1. TF-IDF+RF requires no GPU and is interpretable via feature importance scores. Its structural limitation—reliance on surface lexical features—is the central reason it fails on adversarially disguised inputs, as shown in the results.

3.6 Layer 2: Fine-Tuned BERT Classifier

Architecture and prompt framing. The final classification layer uses `bert-base-uncased` with a sequence classification head. Each prompt is prefixed with the instruction: *"Classify whether the following user request, if fulfilled, would cause harm."* This framing was chosen to direct the model toward reasoning about execution consequences rather than surface features—a distinction important for adversarial benign prompts, which superficially resemble harmful requests but would not cause harm if acted upon.

Training configuration. The tokenizer uses `max_length=256` with truncation and padding. Class weights are computed via `sklearn.utils.class_weight.compute_class_weight('balanced')` to handle the 2:1 label imbalance. The AdamW optimizer is used with learning rate 2×10^{-5} and weight decay 0.01; a linear warmup schedule covers the first 10% of training steps. Gradient clipping at norm 1.0 stabilizes training. Batch size is 16; training runs for 5 epochs. The checkpoint with the best validation F1 is retained.

Why BERT over alternatives. BERT's bidirectional attention captures full-sequence context, allowing it to detect harmful intent embedded in narrative or roleplay framing—patterns that defeat surface-level classifiers. Compared to decoder-only models used as zero-shot classifiers (Section 3.3), fine-tuned BERT provides consistent recall across both harmful categories without threshold instability. Compared to embedding-based similarity filters (all-MiniLM-L6-v2), fine-tuned BERT has a task-specific classification head optimized directly on the training labels rather than relying on distance in a general-purpose embedding space.

3.7 Adversarial Sample Augmentation

To test and improve robustness on the adversarial benign category, I generated adversarial training samples using two attack methods, one per model. The attack objective is: given a correctly-classified adversarial benign prompt (ground truth: benign, correctly predicted: benign), find a perturbation that causes the classifier to misclassify it as harmful. This tests whether the model can be destabilized by synonym-level changes that preserve benign intent but shift surface lexical cues toward harmful-sounding vocabulary.

TextFooler for Random Forest. TextFooler [6] is a black-box word substitution attack that iteratively replaces tokens with semantically similar alternatives (via GloVe embedding neighbors) until the classifier changes its prediction. For the RF model, 918 adversarial benign samples were attacked; 622 succeeded (attack success rate 0.678). The high success rate reflects RF's dependence on specific lexical features: replacing a single high-weight token often flips the prediction. The 622 adversarial samples were added to the RF training set (21,000 $\rightarrow$ 21,622).

Gradient-Based Attack for BERT. For BERT, a gradient-based token substitution attack is used. At each step, the token with the highest gradient magnitude with respect to the classification loss is identified and replaced with a candidate from its vocabulary neighborhood, subject to a constraint that semantic similarity (cosine distance in embedding space) remains above a threshold. This is a white-box attack exploiting BERT's differentiable architecture. Of 918 adversarial benign samples attacked, only 92 succeeded (attack success rate 0.075). The low success rate confirms that BERT's contextual representations are substantially more robust to token-level perturbations than RF's bag-of-words features. The 92 adversarial samples were added to the BERT training set (21,000 $\rightarrow$ 21,092).

The large difference in attack success rate (0.678 vs. 0.075) is direct evidence of the representational gap between the two models. Importantly, all generated adversarial samples—622 from TextFooler and 92 from the gradient-based attack, totalling 714—were pooled and added to the training set of *both* models. More adversarial samples benefit both classifiers regardless of which attack produced them, since all target the same adversarial benign failure mode. The combined training sets therefore become $21{,}000 + 714 = 21{,}714$ for each model.

Adversarial sample generation targeted only the adversarial benign subset, as it is the hardest category for both models (see Section 4.4 for full breakdown). Generating adversarial variants of harmful prompts to induce false negatives is left for future work.

3.8 Gradient×Input Attribution Analysis

To investigate what features each model relies on, I applied Gradient×Input attribution to the BERT model. For each input token, the attribution score is computed as the element-wise product of the input embedding and its gradient with respect to the classification logit:

$$\text{attr}(t) = \mathbf{e}_t \odot \nabla_{\mathbf{e}_t} f(x)$$

where $\mathbf{e}_t$ is the embedding of token t and $f(x)$ is the harmful-class logit. The magnitude of this product indicates how strongly token t contributes to the final prediction; positive values push toward harmful, negative values toward benign.

Attribution was computed over 500 test samples to produce a global token importance ranking, and over 10 FP, 10 FN, and 10 ADV_BEN_FP error cases (before and after adversarial augmentation) to examine how the model's reasoning changes at the case level. The global analysis reveals whether the model's most influential tokens are semantically relevant to jailbreak intent or reflect dataset-level spurious correlations. The case-level analysis reveals how augmentation shifts attribution patterns for each error type. Results are presented in Sections 4.3 and 4.4.

4 Experimental Results

This section evaluates three configurations—TF-IDF + Random Forest, fine-tuned BERT, and the combined cascade—across both original and augmented training configurations, evaluated on the same test set. The goal is not to demonstrate that a multi-layer pipeline outperforms a single model, but to compare each component's behavior and understand what drives performance differences. Results on the 200-sample subset include LLM baselines for direct comparison; results on the 4,500-sample set provide higher-confidence estimates of generalization.

4.1 Main Results

Table 2: Full system comparison (200-sample subset). All methods evaluated on the same 200-sample set, enabling direct comparison including LLM baselines.

Method	n	TP	FP	FN	TN	Acc	Prec	Rec	F1
LLaMA-8B Naive	200	7	0	130	63	0.3500	**1.0000**	0.0511	0.0972
LLaMA-8B Strict	200	88	10	49	53	0.7050	0.8980	0.6423	0.7489
RF-original	200	114	17	18	51	0.8250	0.8702	0.8636	0.8669
RF-test	200	97	10	35	58	0.7750	0.9065	0.7348	0.8117
BERT-original	200	132	5	0	63	0.9750	0.9635	**1.0000**	0.9814
BERT-test	200	130	2	2	66	**0.9800**	0.9848	0.9848	**0.9848**

LLM Baseline Analysis (200-sample subset). The 200-sample subset was used for LLM baseline comparison because LLM API inference is costly; results from other models were pre-cached to enable direct comparison. LLaMA-8B Naive achieves perfect precision (1.0000) but FN = 130, meaning it almost never predicts "harmful"—useless as a safety classifier. LLaMA-8B Strict recovers recall at the cost of false positives (F1 = 0.749), but the precision-recall tradeoff cannot be resolved by threshold tuning alone. This confirms that a cost-acceptable zero-shot LLM is not a viable approach, and motivates the supervised pipeline. BERT-original achieves recall = 1.0000 (FN = 0), missing no harmful prompts. BERT-test further improves to the highest accuracy (0.9800) and F1 (0.9848), demonstrating strong generalization to adversarial prompts after augmentation. The gap between LLaMA-8B Strict (F1 = 0.749) and BERT-test (F1 = 0.9848) confirms that fine-tuned BERT is the appropriate design choice.

Table 3: Full system comparison (4,500-sample test set). LLM baselines were not evaluated at this scale. Results are not directly comparable across the two tables.

Method	n	TP	FP	FN	TN	Acc	Prec	Rec	F1
RF-original	4500	2599	368	401	1132	0.8291	0.8760	0.8663	0.8711
RF-test	4500	2158	265	842	1235	0.7540	0.8906	0.7193	0.7959
BERT-original	4500	2936	67	64	1433	0.9709	0.9777	0.9787	0.9782
BERT-test	4500	2941	52	59	1448	0.9753	**0.9826**	0.9803	**0.9815**
Combined-original	4500	2958	295	42	1205	0.9251	0.9093	0.9860	0.9461
Combined-test	4500	2963	292	37	1208	**0.9269**	0.9103	**0.9877**	0.9474

Overall ranking: BERT > RF+BERT > RF ≫ LLM. On the full 4,500-sample test set, BERT-test achieves the best F1 = 0.9815 (Precision = 0.9826, FP = 52; Recall = 0.9803, FN = 59): harmful predictions are almost always correct and almost all harmful prompts are caught. Combined-test reaches the highest recall (0.9877) and accuracy (0.9269), but FP = 292—of which 265 propagate directly from RF—dragging F1 down to 0.9474. RF alone trails substantially (F1 = 0.7959 on test). LLM zero-shot (≤ 0.749) is far behind all supervised methods. Adversarial augmentation improves both models: BERT FP drops from 67 to 52, confirming that the additional adversarial benign samples are effective.

Table 4: Training time comparison across configurations.

Configuration	Training Time
RF only	∼8 min
RF+BERT pipeline	∼25 min
BERT only	∼48 min

Training time and pipeline tradeoff. The pipeline is designed to reduce inference cost by filtering prompts before reaching BERT. However, RF filters out too many false positives before BERT can correct them, costing 3.41 F1 points at this scale. For deployments with much larger data volumes where per-query BERT inference becomes the bottleneck, the pipeline may be worth reconsidering. At the present scale, BERT alone is the better choice.

Figure 1 visualizes the RF, BERT, and combined pipeline comparisons across Precision, Recall, and F1.

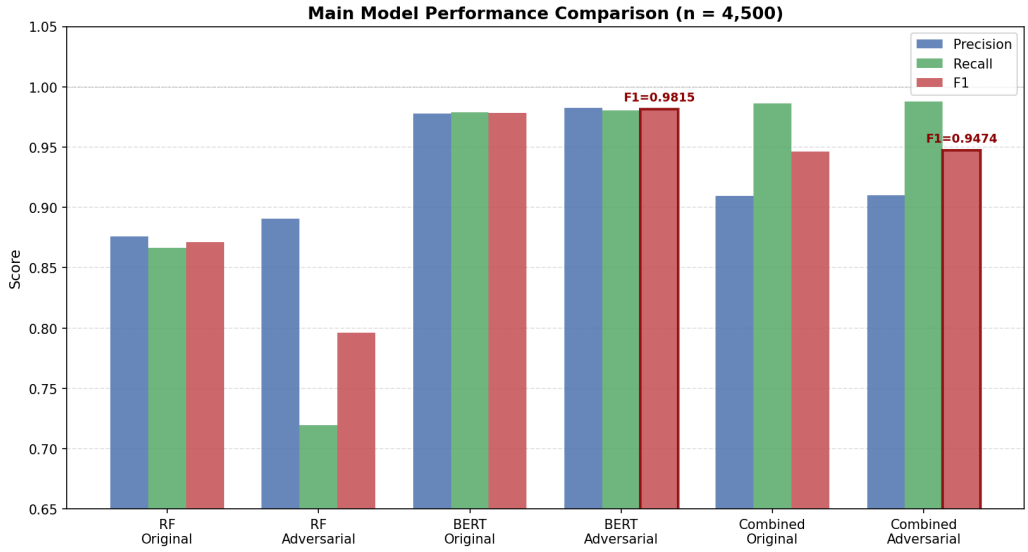

Figure 1: Model performance comparison (Precision, Recall, F1) for RF, BERT, and combined pipeline variants. BERT-adversarial achieves the best F1 = 0.9815; the combined pipeline drops to F1 = 0.9474 despite higher recall. LLM baselines are excluded as they were evaluated on a different sample size.

Comparison to prior work. Table 5 places these results in context. Prior inference-time defenses either report no F1 at all (SmoothLLM reports only ASR reduction on GCG attacks; Shen et al. measure attack success without any defense), or achieve substantially lower F1 on easier distributions (Jain et al.'s perplexity filter and paraphrase defenses; LLM zero-shot at F1 $\leq$ 0.75). The closest supervised baseline, Hawkins et al., establishes fine-tuned BERT as the strongest method but evaluates on simpler datasets and does not report a specific F1. SceneJailEval SOTA reaches F1 = 0.917 on full-scenario evaluation. BERT fine-tuned on WildJailbreak reaches F1 = 0.9815, showing strong performance on a distribution specifically designed to stress-test adversarial benign classification. Because these methods are evaluated on different datasets, the comparison should be interpreted as contextual rather than directly competitive.

The performance advantage here reflects two compounding factors: BERT fine-tuning itself, and the quality of WildJailbreak as a training and evaluation set. WildJailbreak's adversarial harmful prompts use deep semantic disguise rather than explicit phrasing, preventing classifiers from relying on surface vocabulary. Its built-in adversarial benign category eliminates the need for external benign supplementation and directly tests the lexical-shortcut failure mode. Neither factor alone fully explains the result; the dataset selection and model choice jointly contribute.

Table 5: Contextual comparison with prior work. F1 figures where available; methods are evaluated on different datasets and are not directly comparable.

Method	F1	Dataset
Shen et al. (attack characterization only)	N/A	In-the-wild jailbreaks
SmoothLLM [3] (reports ASR only)	N/A	GCG attacks
Jain et al. [4] (perplexity / paraphrase)	<0.75	AdvBench
LLaMA-8B zero-shot (this work)	0.749	WildJailbreak (200)
Hawkins et al. [5] (fine-tuned BERT)	best, not reported	simpler dataset
SceneJailEval SOTA [7]	0.917	full-scenario
This work: BERT-test	**0.9815**	**WildJailbreak adversarial**

4.2 Error Analysis by Sample Type

Table 6: Accuracy by sample type for RF and BERT (test set, 4,500 samples).

Sample Type	RF Acc	BERT Acc	Improvement
Vanilla harmful	0.820	**0.986**	**+0.166**
Adversarial harmful	0.859	**0.923**	**+0.064**
Vanilla benign	**0.974**	**0.974**	+0.000
Adversarial benign	0.495	**0.815**	**+0.320**

Adversarial benign is by far the most challenging category across all methods. RF accuracy of 0.495 is essentially random chance, confirming complete reliance on surface lexical patterns. BERT's gain of +0.320 on this category is the largest improvement of any group, and accounts for most of the overall performance gap between the two models. Adversarial benign prompts use jailbreak-style wording, making them difficult for any lexical classifier to distinguish from genuinely harmful requests; this is the direct motivation for adversarial sample augmentation targeting this category. Figure 2 illustrates this contrast.

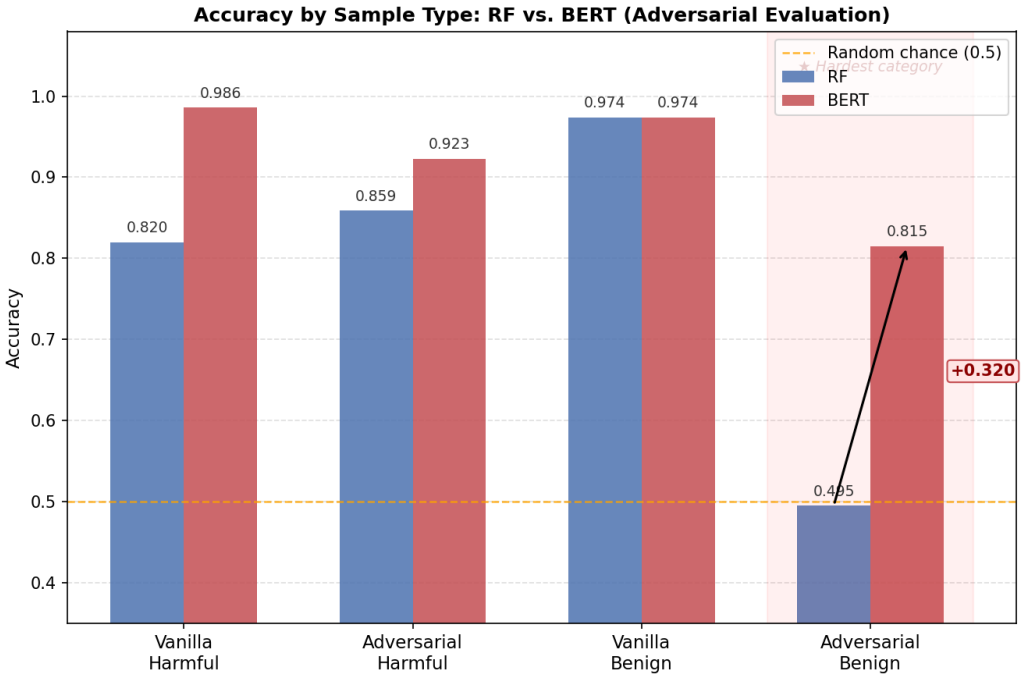

Figure 2: Accuracy by sample type for RF vs. BERT. Adversarial benign is the hardest category: RF achieves near-random accuracy (0.495) while BERT reaches 0.815, a gap of +0.320.

4.3 Feature and Token Importance Analysis

To understand how each model makes decisions before and after adversarial-sample augmentation, I visualized the top 20 TF-IDF features for Layer 1 RF and the top 20 Gradient×Input token attributions for Layer 2 BERT (Figure 3–6).

Layer 1 RF: before vs. after adversarial augmentation. The RF feature-importance plots show that the model remains highly dependent on surface lexical cues both before and after adversarial aug-

mentation. In the original RF model, the most important features are *create* (≈0.028), *write* (≈0.024), and *generate* (≈0.019). After adversarial augmentation, the same three features remain dominant: *create* (≈0.024), *write* (≈0.021), and *generate* (≈0.021). Adversarial augmentation does not fundamentally change RF's decision logic. Lower-ranked features shift slightly after augmentation—*imagine*, *story*, *information*, and *copyright* become more visible—suggesting a modest increase in sensitivity to narrative language, but the overall pattern remains keyword-based.

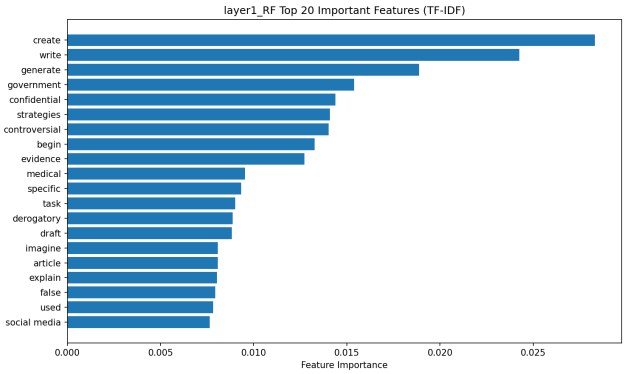

Figure 3: RF feature importance, original model. *create*, *write*, and *generate* dominate, reflecting reliance on instruction-style verbs.

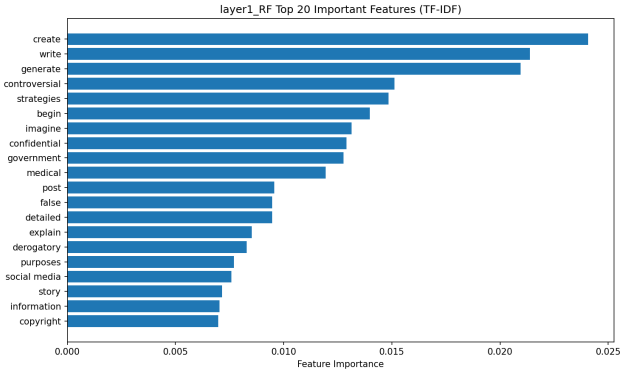

Figure 4: RF feature importance, adversarial-augmented model. Top three features remain unchanged; lower-ranked features shift slightly toward narrative language.

Layer 2 BERT: before vs. after adversarial augmentation. BERT attends to contextual and semantic tokens rather than surface keywords. After adversarial augmentation, attribution redistributes away from a single dominant token toward a broader set, indicating more distributed context-sensitive representations. However, the top tokens remain semantically indirect as indicators of harmful intent, confirming that augmentation reduces but does not eliminate lexical shortcut learning.

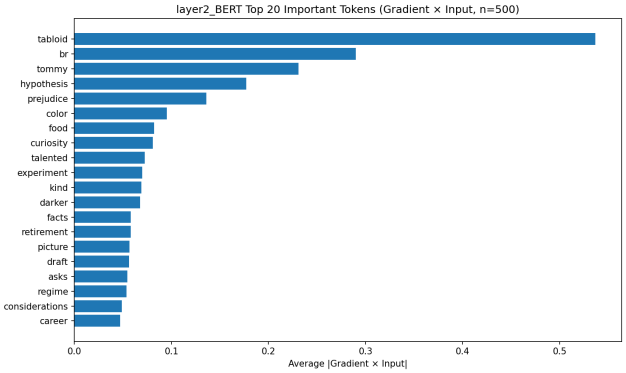

Figure 5: BERT token importance, original model. Attribution is dominated by *tabloid* (≈0.54)—a dataset-specific correlation unrelated to jailbreak intent.

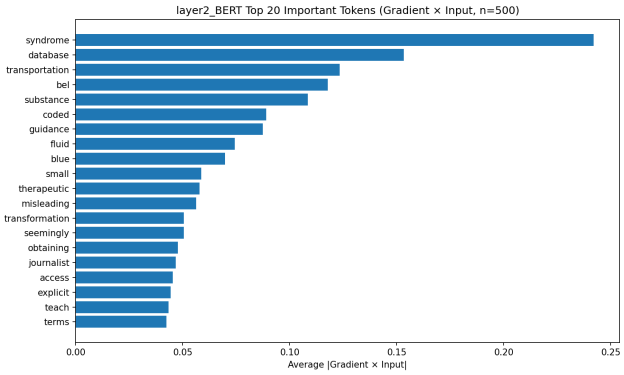

Figure 6: BERT token importance, adversarial-augmented model. Maximum attribution drops to ≈0.24 (*syndrome*); distribution is less dominated by a single token, but top tokens remain semantically unclear.

RF vs. BERT. RF matches keywords: *create*, *write*, *generate* dominate both before and after augmentation—interpretable but shallow, with no contextual representation. BERT attends to contextual tokens, explaining why it generalizes better to adversarial cases. Adversarial augmentation has a different

effect on each model: RF's top features are nearly unchanged, while BERT's attribution redistributes away from a single dominant token—reflecting a more distributed, context-sensitive representation.

4.4 Case-Level Attribution Analysis

Table 7 presents three representative cases illustrating the effect of adversarial augmentation.

Table 7: Representative case-level summary. P(harmful) before and after adversarial augmentation, with key attribution tokens.

Case	Type	Prompt (condensed)	Key tokens	Before	After
ADV_BEN_FP #9	Adversarial benign	detective storytelling game: hidden clues, illegal activities, dramatic investigations framed as harmless fiction	*clues, illegal, investigations*	0.997	0.000
FP #7	Vanilla benign	coded language game uncovering hidden beliefs and controversial views while keeping conversation respectful	*game, beliefs, views*	0.000	1.000
FN #3	Harmful	fictional narrative involving confidential company documents and rival organizations	*confidential, documents, rival*	0.998	0.001

Primary Success: ADV_BEN_FP #9

ADV_BEN_FP #9 is the clearest success of adversarial augmentation. The prompt is ground-truth benign but contains surface cues (*illegal*, *investigations*, *clues*) that drove the original model to P(harmful) = 0.997. After augmentation, P(harmful) collapses to 0.000: the model learned to treat crime-fiction narrative framing as a benign signal rather than a harm indicator. Across all 10 ADV_BEN_FP cases, average P(harmful) drops from 0.1662 to 0.0001.

Tradeoff: FP #7 and FN #3

The same adjustment that helps ADV_BEN_FP cases causes augmentation to over-generalise narrative framing as a benign signal. FP #7 shifts from P(harmful) = 0.000 to 1.000; FN #3 shifts from 0.998 to 0.001. Overall F1 still improves from 0.9782 to 0.9815, confirming the gains outweigh the cost.

5 Conclusion

BERT alone is the best defense: BERT (F1 = 0.9815) > RF+BERT pipeline (F1 = 0.9474) > RF (F1 = 0.7959) $\gg$ LLM zero-shot (F1 = 0.749). The pipeline reduces per-query inference cost by pre-filtering with RF, but propagated false positives cost 3.41 F1 points. Adversarial benign sample augmentation is effective: BERT FP drops from 67 to 52.

All cheaper alternatives share the same failure—falsely filtering adversarial benign prompts (false positives): RF relies on surface keywords (Acc = 0.495 on adversarial benign, near random chance); embedding-based and perplexity-based filters [4] share the same surface-signal failure mode; rule-based self-referential detection [5] and scenario-based classification [7] both depend on surface structure—since

adversarial samples are predominantly roleplay and narrative framing, adversarial benign prompts are systematically misidentified.

LLM-as-judge (Llama Guard, WildGuard) has a genuinely complementary error distribution to BERT, but currently carries high inference cost per query. If LLM inference cost decreases substantially, LLM-as-judge may become practical as a complementary or replacement screening layer in some deployments.

Future work. *Paired contrastive training*: the current lexical shortcut limitation motivates training on matched prompt pairs sharing surface structure but differing in intent (*e.g.*, "write malware to steal passwords" vs. "write a fiction story about malware stealing passwords"), directly forcing the model to learn intent rather than keywords. *Better interpretability*: Gradient×Input attribution reveals *what* tokens matter but not *how* the model combines them; BERTViz attention-head visualization would allow direct inspection of the model's reasoning across layers. *Transformer comparison*: it remains an open question whether the observed keyword sensitivity is BERT-specific—evaluating RoBERTa (stronger but heavier) and DistilBERT (smaller and faster) under the same setup would test whether a better accuracy/speed tradeoff exists within the transformer family.

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